

Hanxiang(Dennis) Hao

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EDUCATION

Georgia Institute of Technology (GT), Atlanta, GA
Bachelor of Science in Electrical Engineering; GPA: 4.0

August 2025 – May 2027

Skills

Technical

- Design Verification, UVM Testbench Architecture, SystemVerilog Assertions, Formal Verification, Functional Coverage, RTL Design/Debugging, FSM/Timing Verification, Pipeline Testing, DUT Functional Validation, Digital Logic Design, Floorplanning/Macro Placement, Static Timing & DRC Analysis, GPU Kernel Optimization, Distributed Training

Tools

- Cadence Xcelium/SimVision/VerisiumDebug/Genus/Innovus, Synopsys VCS/Verdi, Git, Slurm Workload Manager, Verilator, Yosys, SymbiYosys, NVIDIA Nsight Systems/Compute

Programming & Libraries

- C/C++, Python, Java, Verilog/SystemVerilog, Makefile/Shell Scripting, MATLAB, RISC-V assembly, CUDA, PyTorch, NumPy, MPI, OpenMP

EXPERIENCE

Gatech EIC Lab; Machine Learning System Research Intern January 2026 - Present

- Implemented training-free sparse attention with dynamic KV cache eviction (Sparse-dLLM) on Fast-dLLM-v2 (1.5B & 7B), reducing peak memory and improving inference throughput while maintaining accuracy on GSM8K and MATH

Future Computing with Rogues Gallery; Reconfigurable XChaCha FPGA subteam January 2026 – May 2026

- Built a formal verification flow for an open-source XChaCha20 FPGA accelerator (tiny-xcc) using SymbiYosys and OSS CAD Suite; authored SystemVerilog Assertions to prove datapath equivalence to RFC 8439, deterministic 21-cycle latency, and AXI4-Lite/Full interface correctness, with GitHub Actions CI/CD running proofs on every push

SiliconJackets; Design Verification Engineer

August 2025 - Present

- Stress-tested the RV32I CPU pipeline by authoring memory-saturating assembly programs (quicksort, matrix multiplication, convolution, recursive subtree-sum) that fully utilize Tapeout 1's 8KB on-chip SRAM, and integrating riscv-dv—an open-source UVM-based constrained-random RISC-V instruction generator—into the testbench flow to extend coverage beyond directed tests

Supercomputing @ GT; High Performance Computing (HPC) Student Researcher

August 2025 - Present

- Optimized CUDA vector-reduction kernels on Georgia Tech's PACE GPU nodes across 4 implementations (global atomics, tree reduction, shared memory, warp-shuffle); profiled with NVIDIA Nsight Compute to eliminate atomic contention
- Trained a CIFAR-10 CNN classifier in PyTorch across four multi-GPU paradigms: single-GPU baseline, Distributed Data Parallel (DDP), pipeline parallelism, and tensor parallelism; open-sourced pretrained weights, Kaggle Jupyter notebooks, and a four-method comparison as a hands-on workshop

PROJECTS

64-bit SystemVerilog Calculator — RTL Design & Verification Project

- Designed a 7-state FSM RTL calculator in SystemVerilog integrating dual sky130 2KB SRAM banks, a 32-bit ripple-carry adder, and a result buffer with configurable memory-mapped address windows
- Built a layered SV testbench (driver/sequencer/monitor/scoreboard) with directed, randomized, and reset-injection tests across all FSM states; wrote 3 SVA assertions and achieved 99% functional coverage in Cadence Xcelium/Verdi

LICENSES & CERTIFICATIONS

SystemVerilog Accelerated Verification with UVM Training — Cadence

November 2025

- Completed 32-hour professional training on UVM 1.2-based verification using Xcelium 2103, covering UVC creation, sequences, TLM, scoreboards, and register modeling

Deep Learning Techniques in MATLAB for Image Applications — MathWorks

May 2026

- Completed a learning path covering CNN architectures, training hyperparameter tuning, regression with deep learning, and YOLO-based object detection